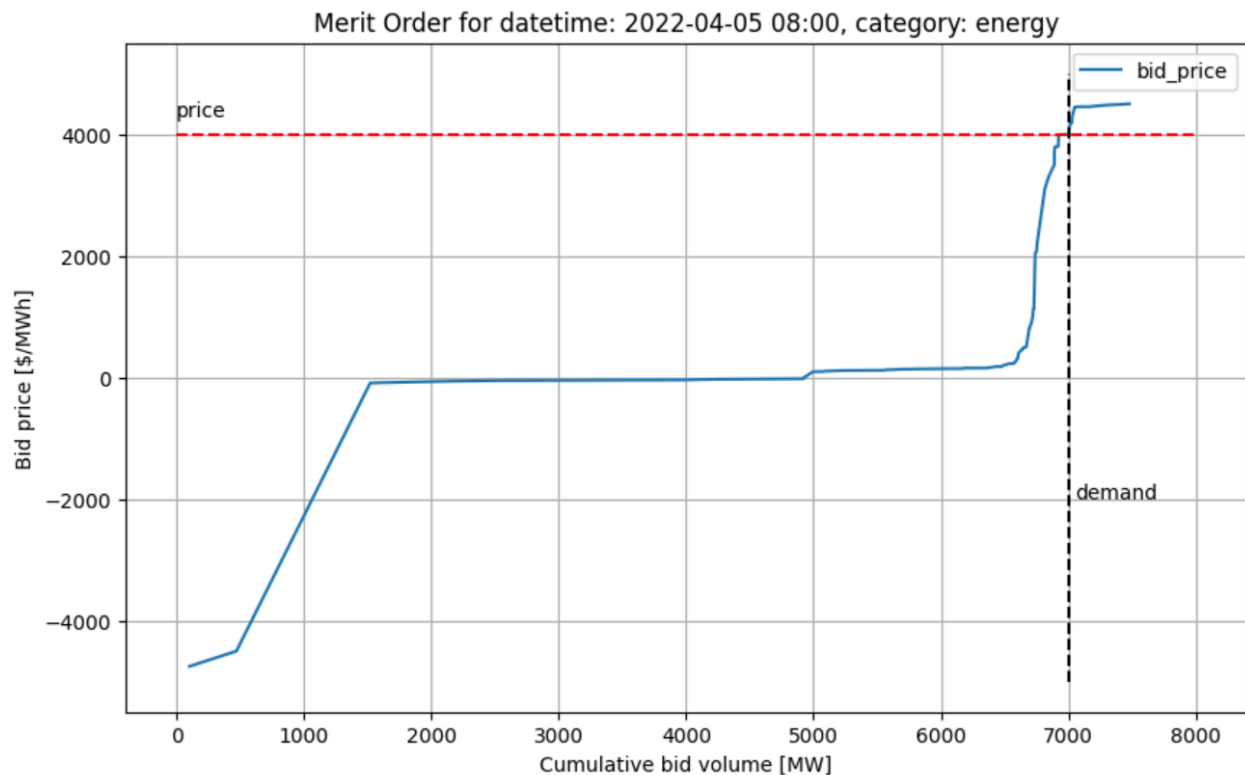


Data Analyst - Pricing & Revenue Assurance

Introduction

The Uniform Singapore Energy Price (USEP) is the wholesale electricity price in Singapore. This price is formed by a so-called market clearing mechanism, in this market the supply and demand are 'matched'. All suppliers submit their offers (how much electricity they're willing to produce, at what price) to the market. All these offers are then ordered from the cheapest to the most expensive offers. The volumes are all added together to get a cumulative volume, price curve, this curve is called the 'merit order' curve.

An example of a merit order curve is shown in the figure below:



We can see that the final USEP price, the wholesale electricity price that all consumers would pay, would be equal to 4000 \$/MWh, if the demand was 7000 MW in this specific period (one period is 30 minutes). If the demand was 4000 MW, the price would be equal to approximately 0 \$/MWh, apparently these units want to generate energy for free! This example illustrates that the final energy price is always set by the most expensive generator that needs to be activated.

This merit order data is published by the market authority and we can use it to construct estimations of the USEP price, based on the merit order and the demand per period, that is what we'll do in this assignment. Period 1 is from 00:00 till 00:30, period 2 is from 00:30 until 01:00 etc.

Input data

- Merit order: "DelayedOfferStacks_Energy_01-Jan-2023 to 31-Jan-2023.csv"
- Demand [DEMAND] & final prices [USEP]: "USEP_Jan-2023.csv"

Assignment

Preferably the analysis would be performed in a Jupyter notebook or in a python script. Please also make sure to include markdown blocks (you can use comments if you use a python script) that explain what you did. The idea would be to work on this assignment for max 1-3 hours, the main goal would be to understand what your way of working is and to get some insight into the thought process and steps that you take to tackle the problems at hand.

These first questions provided in this assignment are mainly to test some data exploration & data modeling techniques. During the interview a few more tasks will be asked to the interviewee to continue the assignment to further test your reasoning and analytical skills. Make sure that you bring your laptop with your coding setup during the interview (AI coding agents are allowed) so that we can see how you'd implement the answers to some of the questions in real-life.

Tasks

1. Write some code to clean the merit order and demand file, also write down any special findings (please note that there might be data gaps or outliers!). Make sure that all columns match the following lowercase formatting: 'demand', 'bid_volume' etc. Also write a function that will create a datetime index, based on the date and period columns.
2. Plot the merit order for one day, with the cumulative bid volume, it should approximately look like the example above.
3. Write a simple function that takes the demand as an input and return the final clearing price as an output.
 - Bonus points: Write a unit test for your function

Bonus tasks

1. Write and store (in the notebook/python file) some code to spin up a database to perform some SQL analysis on the data. We recommend using DuckDB, but you are free to choose any database tool (e.g. SQLite, Docker container hosting PostgreSQL etc).
2. Perform the following SQL tasks for your database:
 - a. Load the merit order and demand files into tables from the original provided files.
 - b. Perform similar cleaning as task 1 for the tables but in SQL.